

Applied Statistics Exam

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16-17/1 - 2026

1 Distributions and probabilities

1.1

1.1.1

The problem is binomial with $n_1 = 20, p_1 = 0.5$ and $n_2 = 2500, p_2 = 0.004$. I simulated 10^6 trials and also calculated analytically. Results are shown in table 1. The uncertainties are from the error on a fraction.

Method	$P(N_1 > N_2)$	$P(N_2 > N_1)$
Simulation	0.4596 ± 0.0005	0.4373 ± 0.0005
Analytical	0.4597	0.4374

Table 1: Results from simulated trials in problem 1.1.1.

1.1.2

Once again simulated 10^6 trials and also calculated analytically. Errors still from error on a fraction and results are shown in table 3.

Method	$P(25 \leq N_1 \leq 250)$	$P(25 \leq N_2 \leq 250)$
Simulation	$< 10^{-20}$	0.000047 ± 0.000007
Analytical	$< 10^{-20}$	0.000017

Table 2: Results for problem 1.1.2.

1.1.3

Once again simulated 10^6 trials and also calculated analytically. Errors still from error on a fraction and results are shown in table 3.

Method	$P(N_1 = 10)$	$P(N_1 = 20)$	$P(N_2 = 10)$	$P(N_2 = 20)$
Simulation	0.1758 ± 0.0004	0.0000010 ± 0.0000010	0.1260 ± 0.0003	0.00188 ± 0.00004
Analytical	0.17620	0.0000010	0.12536	0.00184

Table 3: Results for problem 1.1.2.

2 Error propagation

2.1

2.1.1

Errors from x and y are propagated to z using the following formula obtained with sympy

$$\sigma_z = \sqrt{\sigma_x^2 \left(-\frac{2x \sin(x^2)}{\log(xy)} - \frac{\cos(x^2)}{x \log(xy)^2} \right)^2 + \frac{\sigma_y^2 \cos^2(x^2)}{y^2 \log(xy)^4}}, \quad (1)$$

using this you get

$$z = -0.343 \pm 0.010. \quad (2)$$

An important thing to note is that the errors are not gaussian. This can be seen in fig. 1. Here 10^5 samples are drawn uniformly from x and y using their value and uncertainty. z is then calculated for all values. The gaussian sample and pdf uses the central value for z as mean and its uncertainty σ_z as scale.

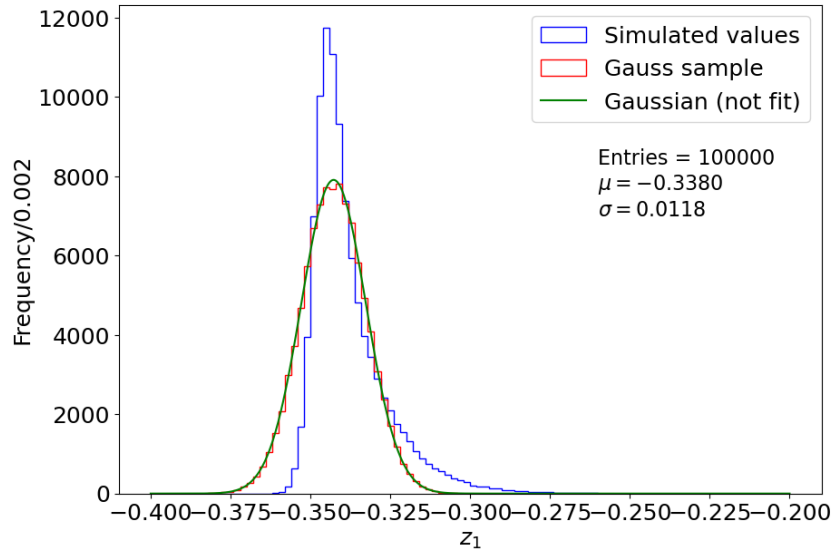


Figure 1: Non-gaussian errors for 2.1.1. The gaussian sample and pdf the central value for z as mean and its uncertainty σ_z as scale.

2.1.2

In the correlated case with $\rho_{xy} = 0.87$, the error propagation changes to

$$\sigma_z = \left[-\frac{2\rho_{xy}\sigma_x\sigma_y \cos(x^2)}{y \log(xy)^2} \left(-\frac{2x \sin(x^2)}{\log(xy)} - \frac{\cos(x^2)}{x \log(xy)^2} \right) + \sigma_x^2 \left(-\frac{2x \sin(x^2)}{\log(xy)} - \frac{\cos(x^2)}{x \log(xy)^2} \right)^2 + \frac{\sigma_y^2 \cos^2(x^2)}{y^2 \log(xy)^4} \right]^{1/2} \quad (3)$$

using this you get

$$z = -0.343 \pm 0.007. \quad (4)$$

2.2

2.2.1

The mean density value and its uncertainty for each temperature is shown in table 4. The error comes from standard error on the mean (SEM). Since this is a sample the std used for the error utilizes Bessel correction ie. $ddof = 1$.

$T = 0.2$	$T = 2.3$	$T = 3.9$
$(999855 \pm 3) \times 10^{-6}$	$(999952 \pm 3) \times 10^{-6}$	$(999972 \pm 3) \times 10^{-6}$

$T = 5.9$	$T = 7.8$
$(999951 \pm 4) \times 10^{-6}$	$(999865 \pm 4) \times 10^{-6}$

Table 4: Results from problem 2.2.1.

2.2.2

Chauvenets filter is used to exclude measurements. The amount of rejected measurements, their value and z-value are all shown in fig. 2. The new density means and uncertainties are shown in table 5.

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For T= 0.20: Accepted points = 14, Rejected points = 6
Rejected points = [0.9998939 0.9998328 0.999832 0.9998672 0.9998468 0.9998508]
with z-values = [3.27070846 2.5365456 2.63622371 2.964003 2.248554 2.58277288]

For T= 2.30: Accepted points = 17, Rejected points = 3
Rejected points = [0.9999116 0.999996 0.9999576]
with z-values = [3.00819561 3.22610051 2.31864913]

For T= 3.90: Accepted points = 18, Rejected points = 2
Rejected points = [0.9999182 0.9999697]
with z-values = [4.30879332 2.6768085]

For T= 5.90: Accepted points = 16, Rejected points = 4
Rejected points = [1.0000143 0.9999809 0.9999651 0.9999263]
with z-values = [3.6016827 3.25656979 2.84662169 2.98181744]

For T= 7.80: Accepted points = 16, Rejected points = 4
Rejected points = [0.9998054 0.9999117 0.9998969 0.9998681]
with z-values = [3.22095494 2.53007403 4.01198979 2.32300375]

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Figure 2: Chauvenet filter used to exclude outliers.

$T = 0.2$	$T = 2.3$	$T = 3.9$
$(9998557 \pm 4) \times 10^{-7}$	$(9999518 \pm 5) \times 10^{-7}$	$(9999752 \pm 4) \times 10^{-7}$

$T = 5.9$	$T = 7.8$
$(9999462 \pm 4) \times 10^{-7}$	$(9998635 \pm 4) \times 10^{-7}$

Table 5: Results from problem 2.2.2 after using Chauvenet filter.

2.2.3

In fig. 3 the means of the water density and their uncertainties are plotted. The density at $T = 3.9$ seems to be the highest with the densities at $T = 2.3$ and $T = 5.9$ being the closest. The z-values and corresponding two-sided p -values are shown in table 4. Since the p -values for both $T = 2.3$ and $T = 5.9$ is $< 10^{-20}$ the numbers are obviously not consistent with each other and water has the highest density at $T = 3.9$.

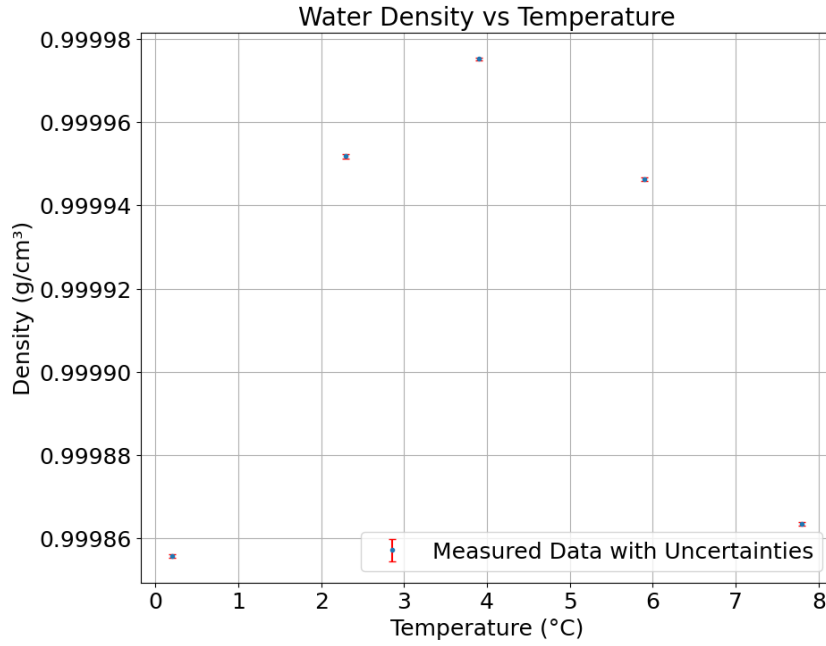


Figure 3: Means of water density at different temperatures.

Value	$T = 2.3$ and $T = 3.9$	$T = 3.9$ and $T = 5.9$
z	37.39	50.75
p	$< 10^{-20}$	$< 10^{-20}$

Table 6: z - and p -values comparing the density of the water $T = 3.9$ with $T = 2.3$ and $T = 5.9$.

2.2.4

I fitted with both a quadratic function and a gaussian + constant. The fits are shown in fig. 4. They have very low p -values ($p < 10^{-5}$ for both), this is in part due to the very small uncertainties on the density measurements. For the G + const. the temperature at which the water has the highest density is the fit parameter $\mu \pm \sigma_\mu$. For the quadratic fit it is the x -value of its vertex given by

$$T_x = \frac{-b}{2a}, \quad (5)$$

with uncertainties from error propagation as

$$\sigma_{T_x} = \sqrt{\left(\frac{\sigma_b}{2a}\right)^2 + \left(\frac{b\sigma_a}{2a^2}\right)^2}. \quad (6)$$

To test if these are consistent a z and two-sided p -value was calculated. These along with the temperatures at which water has the highest density are listed table 7. As can be seen from the p -value the two values are consistent with each other on a 95% confidence interval. The quadratic fit is much better so I would choose that value if need be.

T_{max} (Quadratic)	T_{max} (Gauss)	z	p
4.06 ± 0.02	4.061 ± 0.005	0.26	0.79

Table 7: Temperatures at which water has the highest density. To compare z - and p -value is also calculated.

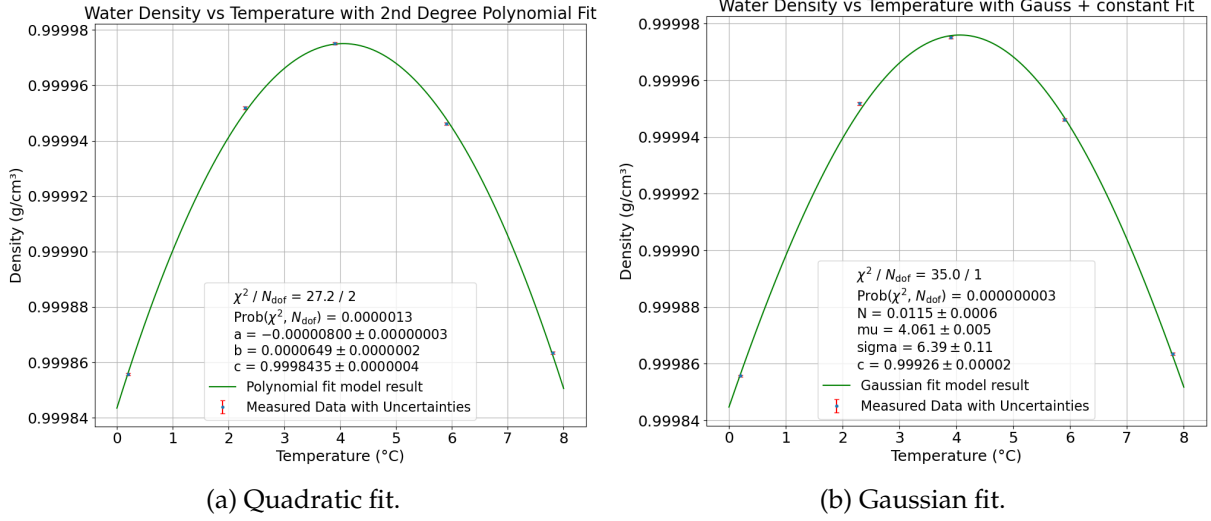


Figure 4: The two fits to the density of water as a function of temperature. For the polynomial fit $P(\chi^2, N_{dof}) = 1 \times 10^{-6}$ and for the gaussian $P(\chi^2, N_{dof}) = 3 \times 10^{-9}$.

2.3

2.3.1

Errors from β is propagated to γ using the following formula obtained with sympy

$$\sigma_\gamma = \sqrt{\frac{\beta^2 \sigma_\beta^2}{(1 - \beta^2)^3}}, \quad (7)$$

yielding

$$\gamma = 1.154 \pm 0.015. \quad (8)$$

2.3.2

For $\beta = 0.95 \pm 0.02$ the value obtained is

$$\gamma = 3.2 \pm 0.6. \quad (9)$$

The uncertainty is not symmetric. This can be seen in fig. 5. Here 10^5 samples are drawn uniformly from β and using its value and uncertainty. γ is then calculated for all values. The gaussian sample and pdf uses the central value for γ as mean and its uncertainty σ_γ as scale. When calculating the upper and lower bound of β you get that are not equally far from γ

$$\gamma_u(\beta = 0.97) = 4.1 \text{ and } \gamma_l(\beta = 0.93) = 2.7. \quad (10)$$

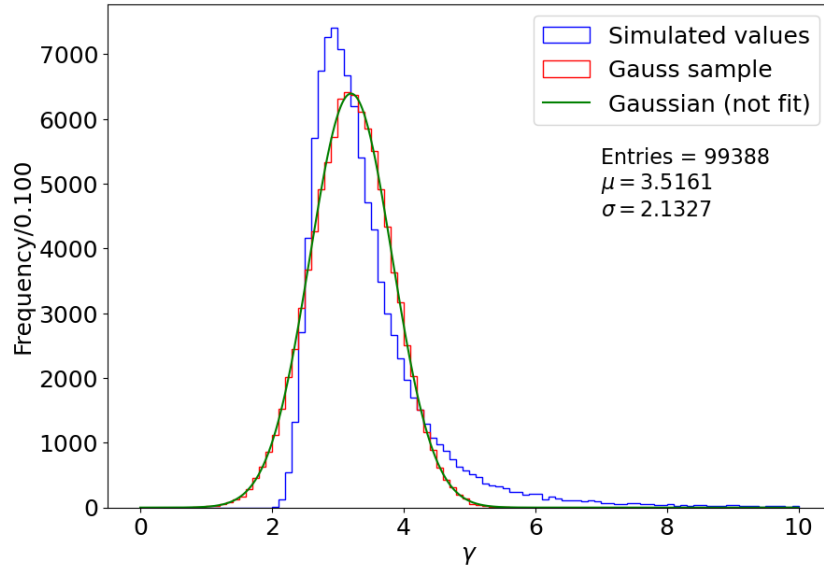


Figure 5: Non-gaussian errors for 2.3.2. The gaussian sample and pdf uses the central value for γ as mean and its uncertainty σ_γ as scale.

3 Simulation / Monte Carlo

3.1

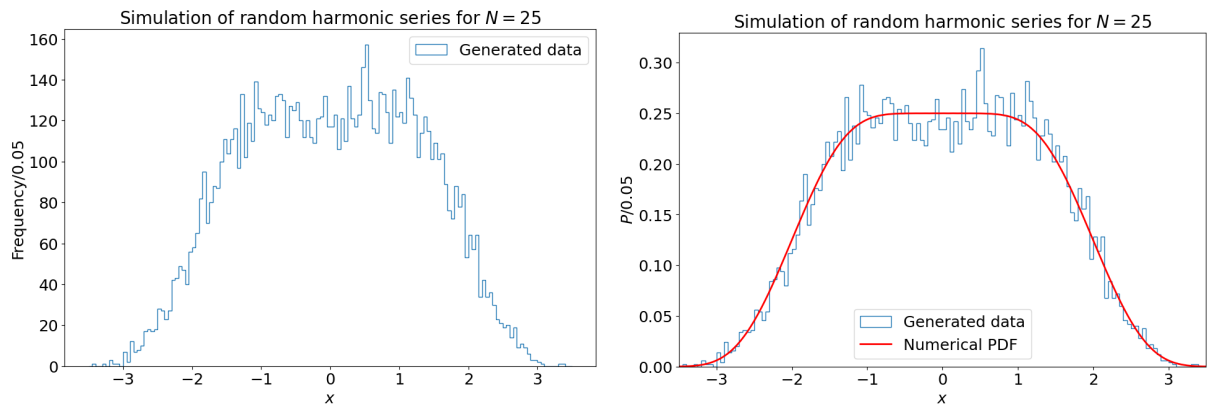
3.1.1

In fig. 6a the distribution of 10000 values of x for $N = 25$ is shown.

The random harmonic series does not have a closed form PDF. It does however have a characteristic equation

$$\varphi = \prod_{n=1}^N \cos(t/n). \quad (11)$$

Taking the inverse Fourier transform of this it can be numerically evaluated. It is plotted in fig. 6b.



(a) Distribution of 10000 simulated values of x for $N = 25$. (b) Numerically evaluated PDF plotted on top of simulated x -values.

Figure 6: Simulated values of x and true numerical PDF.

3.1.2

To test if the distribution of x is symmetric around 0 the numerical value of $x < 0$ is plotted alongside the values of $x > 0$. A two-sample Kolmogorov-Smirnov test is performed yielding $p = 0.967$. This is not sufficiently low to reject the hypothesis that x is not symmetric around 0.

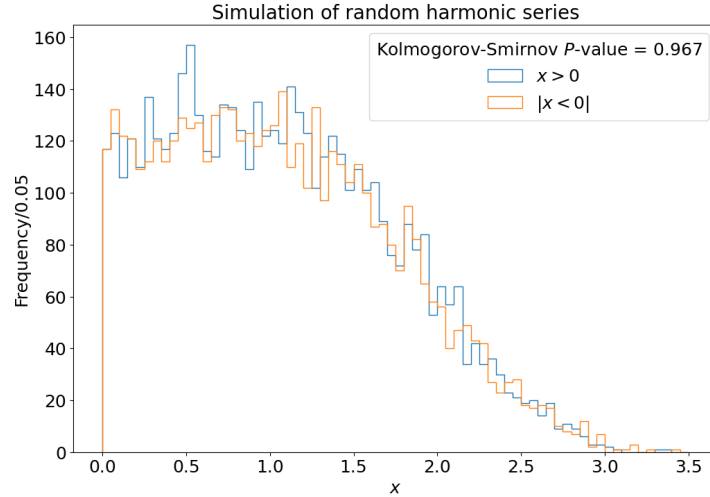
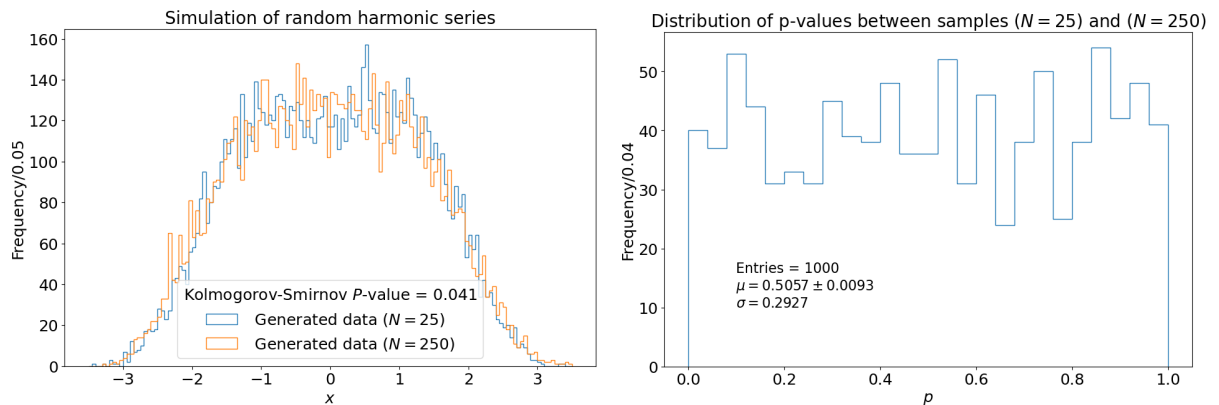


Figure 7: Distribution of $|x < 0|$ plotted alongside $x > 0$. A two-sample KS test yields $p = 0.967$.

3.1.3

10000 values of x for $N = 250$ is plotted alongside the previously simulated 10000 values for $N = 25$ fig. [8a](#). Once again a two-sample Kolmogorov-Smirnov test is performed yielding in this case $p = 0.041$. Individual KS tests sometimes reject the null hypothesis due to sampling variability. Repeated simulations fig. [8b](#) show that the distribution of p -values is uniform on $[0, 1]$ as expected. This indicates that the distributions for $N = 25$ and $N = 250$ are statically consistent.



(a) 10000 values of x for $N = 250$ is plotted alongside the previously simulated 10000 values for $N = 25$. A two-sample KS test yields $p = 0.041$. (b) Distribution of p -values from two-sample KS test between distribution for $N = 25$ and $N = 250$. Uniform on $[0, 1]$ as expected.

Figure 8: Distribution of x for $N = 250$ and distribution of p -values.

3.1.4

I had several different methods for testing if the maximum PDF value of x was consistent with $1/4$. The first method included simulating 10000 different distribution of x (10000 points) for both $N = 25$ and $N = 250$ and then extracting the maximum y -value of a normalized histogram while the second method consisted of using scipys kernel density estimation (KDE) to estimate a PDF and then extract its maximum y -value. In fig. 10 KDE is plotted alongside the distributions of x for $N = 25$ and $N = 250$. Comparing it to fig. 6b it is quite far from the true PDF with its bend in the middle.

Binwidth plays quite an important role in the first method as shown by fig. 9. The maximum y -value varies with different binwidth. This is not the case for KDE.

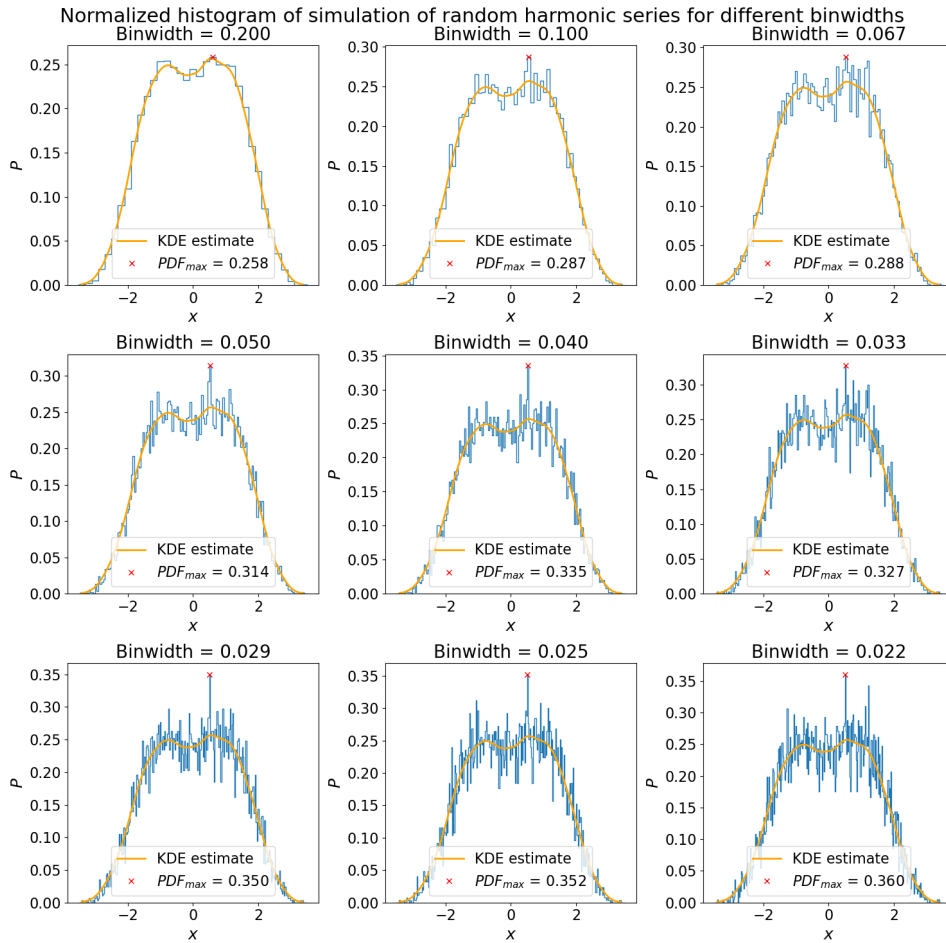


Figure 9: The same distribution of x for $N = 25$ with varying binwidth. The maximum y -value and KDE estimate are also plotted.

I performed 10000 trials where I each time simulated 10000 values of x for both $N = 25$ and $N = 250$. I extracted the maximum y -value of the PDF (density=True) distribution for binwidth= 0.05 and binwidth= 0.2. For the second method I only performed 1000 trials as it was slower than the other two methods. The methods are shown in fig. 11, 12, and 13.

The means μ from the simulation of the maximum PDF value alongside z - and p -values are listed in table 8 and table 9. The maximum y -value of the numerical true PDF is also used. It does however not depend on the simulated values of x . There is no uncertainty on the true PDF max-value and therefore no z - and p -value.

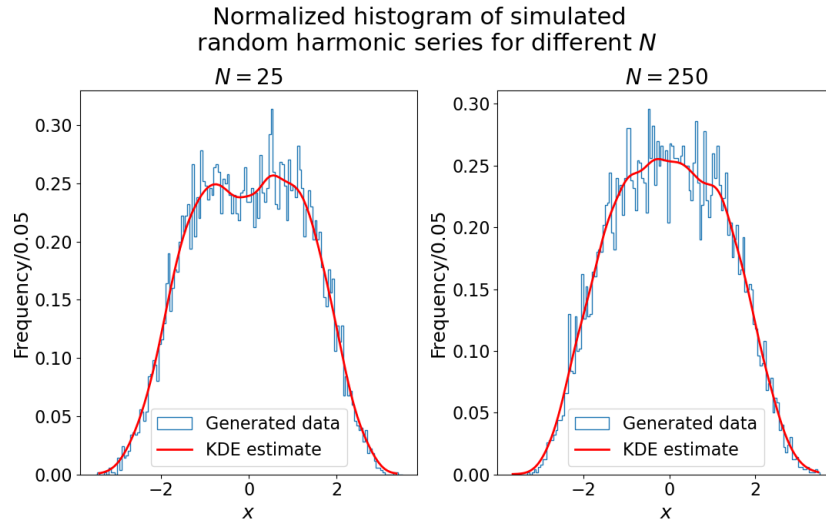


Figure 10: Scipy KDE plotted alongside the distributions of x for $N = 25$ and $N = 250$.

For all methods $p < 10^{-20}$. So the maximum PDF value of x is not consistent with $1/4$ given my methods.

The true numerical PDF has a maximum value that is only 0.02% away from $1/4$. I also found the maximum PDF value for larger binwidths and the best result I got was for $N = 250$ and a binwidth of 1. Still the value was 44.33 sigmas away. If you wanted the maximum value to converge towards the correct maximum of $1/4$ you should instead increase the truncation N and include more terms.

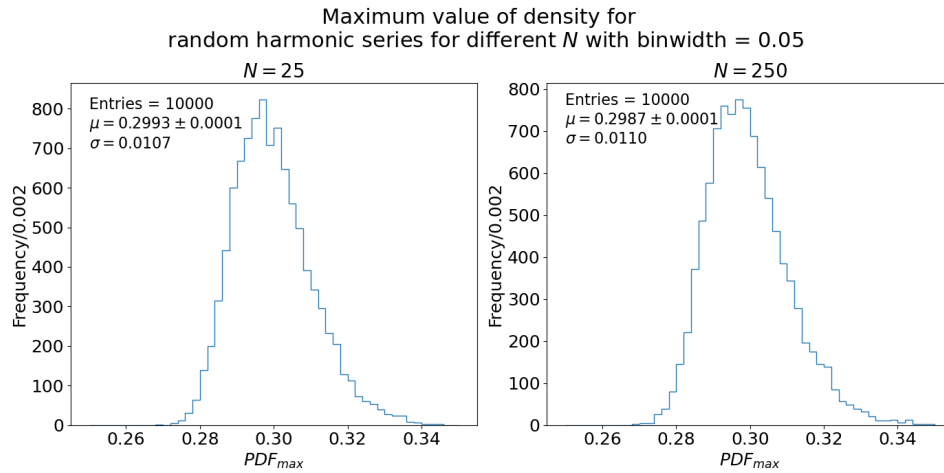
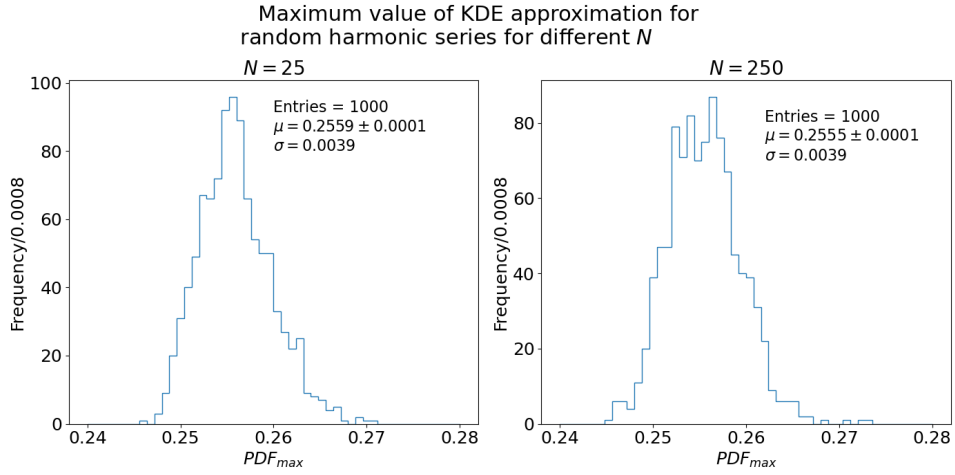
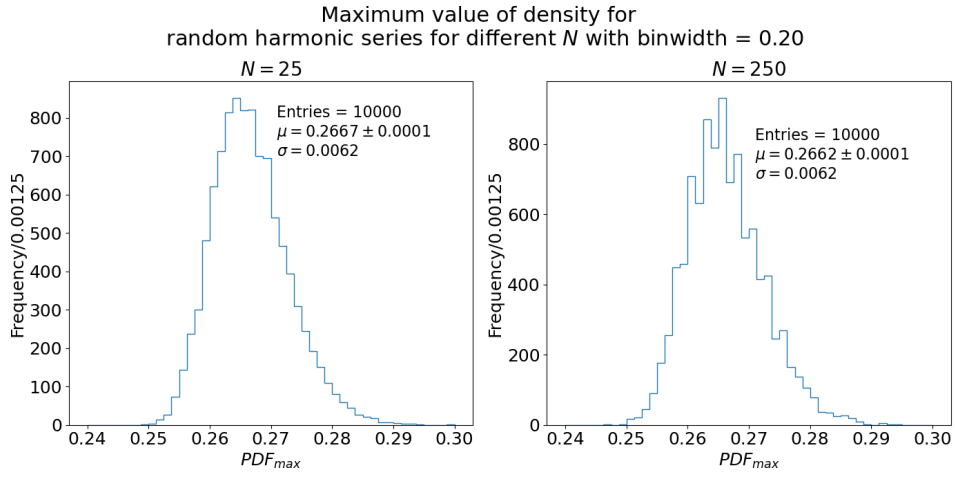


Figure 11: Distribution of extracted PDF_{max} values for binwidth= 0.05.



Value	Binwidth = 0.05		Binwidth = 0.2	
	$N = 25$	$N = 250$	$N = 25$	$N = 250$
μ	0.29933 ± 0.00011	0.29869 ± 0.00011	0.26669 ± 0.00007	0.26618 ± 0.00006
z	459.13	442.05	270.86	259.75
p	$< 10^{-20}$	$< 10^{-20}$	$< 10^{-20}$	$< 10^{-20}$

Table 8: Means μ alongside z - and p -values of the maximum PDF value for different methods (part 1).

Value	KDE		PDF
	$N = 25$	$N = 250$	
μ	0.25593 ± 0.00012	0.25549 ± 0.00012	0.25005
z	48.49	44.74	—
p	$< 10^{-20}$	$< 10^{-20}$	—

Table 9: Means μ alongside z - and p -values of the maximum PDF value for different methods (part 2).

4 Statistical tests

4.1

4.1.1

The mean SBP for healthy patients is

$$\bar{s} = 120.7 \pm 0.3, \quad (12)$$

with the uncertainty from the standard error on the mean. The 95% confidence interval is calculated as

$$\mu \pm 1.96\sigma, \quad (13)$$

with σ being the standard deviation of the sample of SBP from healthy patients.

$$CI_{95\%} = [101.0, 140.3]. \quad (14)$$

4.1.2

The ROC curve is shown in fig. 14.

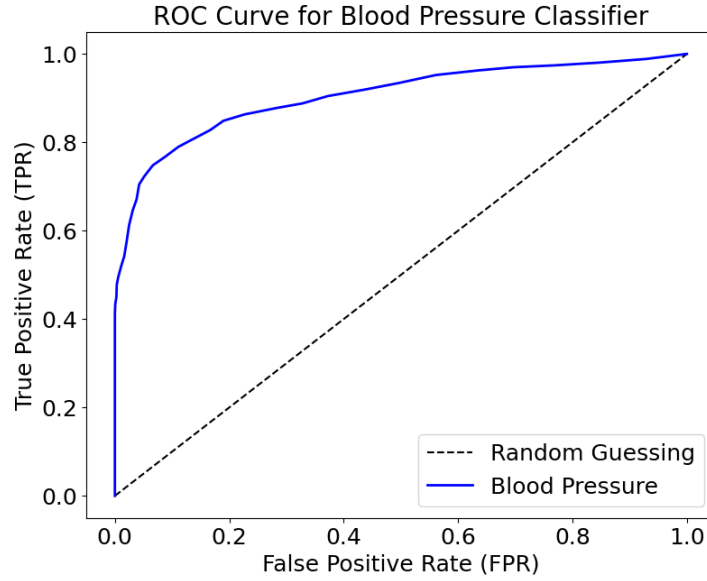


Figure 14: Roc curve for test statistic $d = |SBP - \bar{s}|$.

4.1.3

Using Bayes' theorem

$$P(H_2|SBP = 98) = \frac{P(SBP = 98|H_2)P(H_2)}{P(SBP = 98)}, \quad (15)$$

$P(SBP = 98|H_2)$ can be obtained by calculation directly from the data assuming gaussian data and then extracted the mean and standard deviation. The SBP for H_2 patients alongside a gauss is shown in fig. 15. $P(SBP = 98)$ can be calculated directly by

$$\begin{aligned} P(SBP = 98) &= P(SBP = 98|H_2)P(H_2) + P(SBP = 98|H_0)P(H_0) \\ &= P(H_2)(P(SBP = 98|H_2) + P(SBP = 98|H_0)), \end{aligned} \quad (16)$$

the final step uses $P(H_2) = P(H_0)$. Substituting eq. (16) into eq. (15) the following equation is obtained

$$P(H_2|SBP = 98) = \frac{P(SBP = 98|H_2)}{P(SBP = 98|H_2) + P(SBP = 98|H_0)}. \quad (17)$$

$$P(SBP = 98|H_2) = 0.06145 \text{ and } P(SBP = 98|H_0) = 0.00311, \quad (18)$$

resulting in

$$P(H_2|SBP = 98) = 0.95180. \quad (19)$$

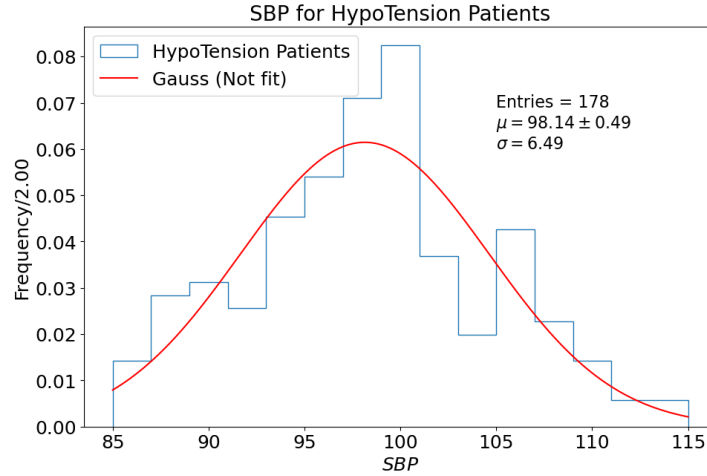


Figure 15: Distribution of SBP for H_2 patients alongside a gauss (not fitted). The mean μ and std σ are also listed.

4.1.4

Using priors obtained from the data yields

$$P(H_0) = 0.61369 \text{ and } P(H_2) = 0.07125, \quad (20)$$

thus $P(H_0) \neq P(H_2)$ and eq. (17) is no longer valid. Instead

$$P(H_2|SBP = 98) = \frac{P(SBP = 98|H_2)P(H_2)}{P(SBP = 98|H_2)P(H_2) + P(SBP = 98|H_0)P(H_0)} = 0.69632. \quad (21)$$

4.1.5

I tried to fit the distribution of SBP values with a Weibull distribution ($p < 10^{-20}$ and inverse gamma distribution $p < 10^{-10}$). As evident by the p -values neither were any good. The plots are shown in fig. 16. I fitting with both a χ^2 and an unbinned likelihood (ullh) fit.

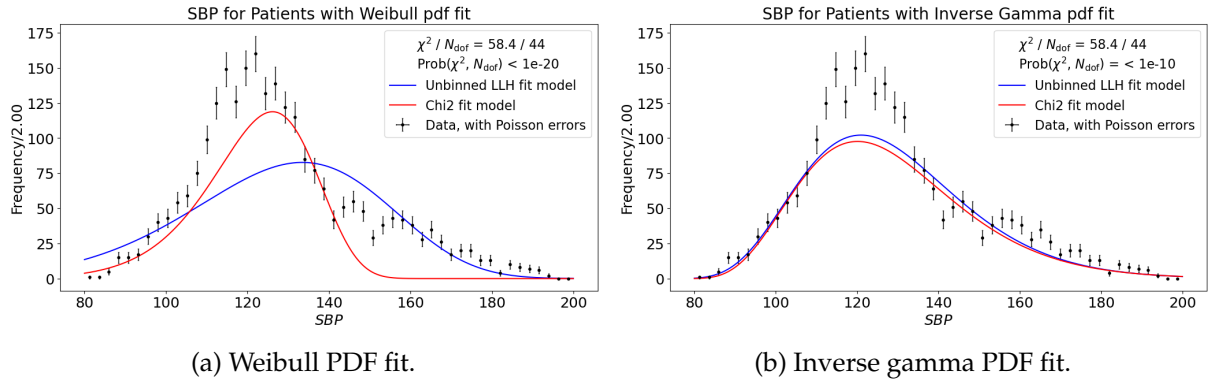


Figure 16: Fits to the distribution of SBP values. Both ullh and χ^2 fits are performed. The p -values are very low indicating a poor fit.

I then fitted, both χ^2 and ullh, a gauss to the SBP values for all three patients groups. Using the extracted parameters from those I fitted a triple gauss to distribution of all SBP values. The fits to the SBP values for each of the patient types are shown in

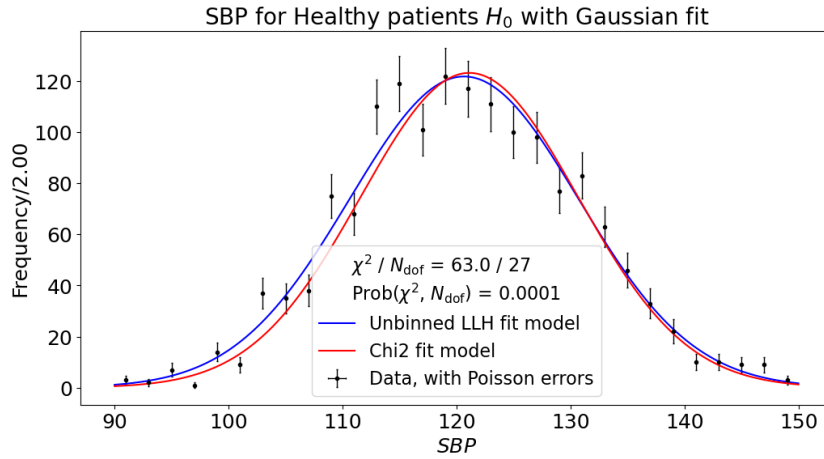


Figure 17: A gauss fitted to SBP values for healthy patients. $p = 0.001$ indicates a poor χ^2 fit. An ullh fit is also shown.

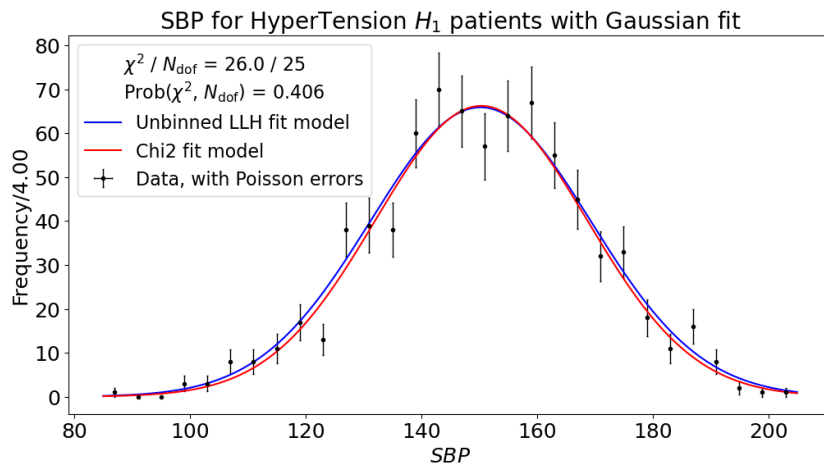


Figure 18: A gauss fitted to SBP values for HyperTension patients. $p = 0.406$ indicates a good χ^2 fit. An ullh fit is also shown.

fig. [17](#) [18](#), and fig. [19](#). For the healthy patients $p = 0.0001$ indicating a poor fit. For

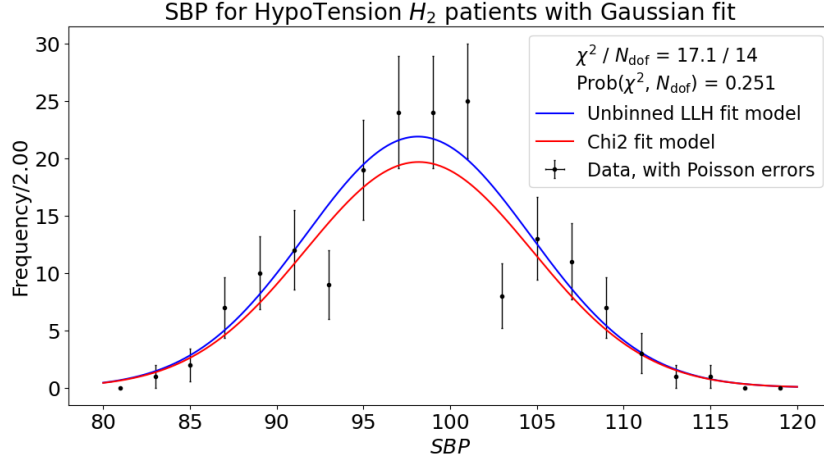


Figure 19: A gauss fitted to SBP values for HypoTension patients. $p = 0.251$ indicates a good χ^2 fit. An ullh fit is also shown.

the other two the fit is good ($p > 0.25$ for both). The triple gauss fit to all SBP values yields $p = 0.023$ means that the fit is rejected at a 95% CI, but accepted at 99%, so it is in between a good and a poor fit.

86% of the bins have more than 5 entries so a χ^2 should be reasonable. We can however see that the ullh fit has a smaller peak around $SBP = 120$.

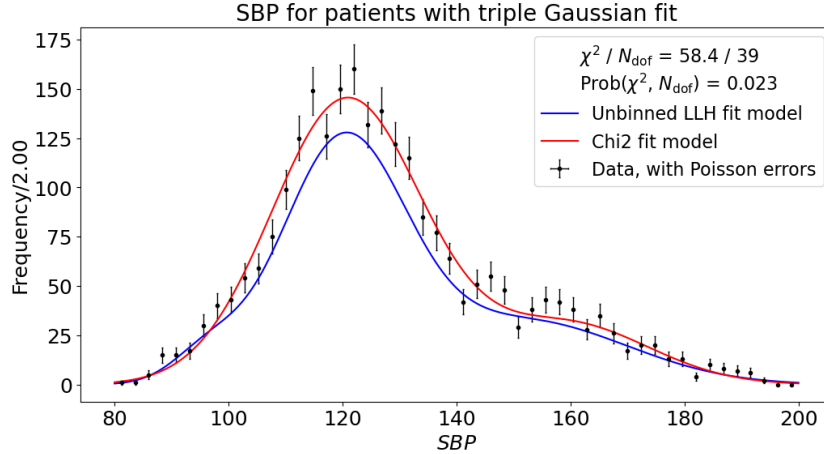


Figure 20: A triple gaussian fitted to SBP values for all patients. $p = 0.0023$ means that the fit is rejected at a 95% CI, but accepted at 99%, so it is in between a good and a poor fit. The ullh fit has a smaller peak around $SBP = 120$.

In order to get a goodness of fit measure for the ullh fit I resampled new SBP values. I did this by sampling from three gaussians using the ullh fit parameters obtained from the fit to the SBP values. I resampled and fitted 1000 times and extracted the minimized LLH value for each trial. I then plotted this distribution of LLH values and calculated a p -value for the ullh fit performed in fig. 20. I calculated this as

$$p = \frac{\#(LLH_{\text{sim}} > LLH_{\text{data}})}{\#(LLH_{\text{sim}})} = 0.733, \quad (22)$$

where LLH_{data} is the value obtained from the fit to the original data. This p -value is not sufficiently low to reject the hypothesis that an ullh fit of a triple gaussian is good model in describing the SBP values of all patients. The distribution of LLH values is shown in fig. 21.

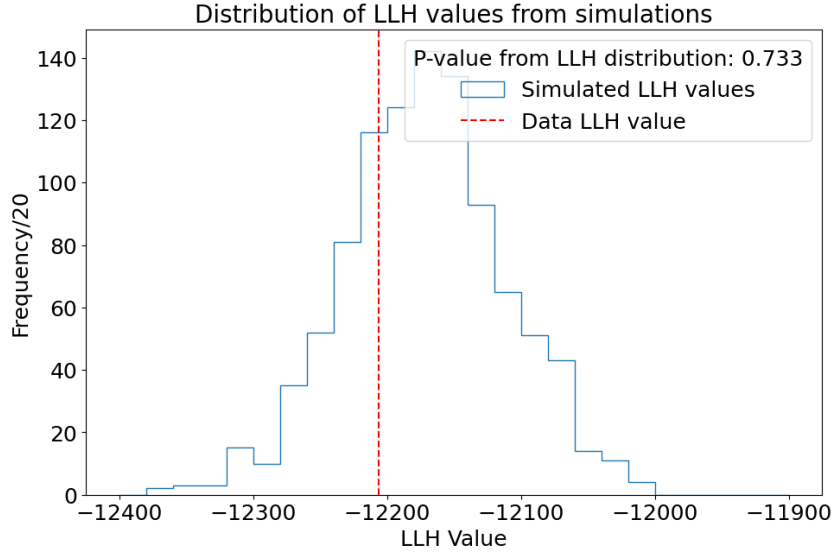


Figure 21: Distribution of LLh values for simulated SBP values reusing the extracted parameters obtained from the ullh fit to the original SBP data.

5 Fitting data

5.1

5.1.1

The uncertainties on t are converted to uncertainties on d as

$$\sigma_{d,i} = v\sigma_{t,i}, \quad (23)$$

where v is the slope of the fitted function. Since this error propagation uses a fit parameter the process is done iteratively until $\sigma_{d,i}$ converges. The fit and data points with converted uncertainty is shown in fig. 22. For the goodness of the fit $p < 10^{-20}$ which is sufficient to reject the hypothesis that the model describes the data well.

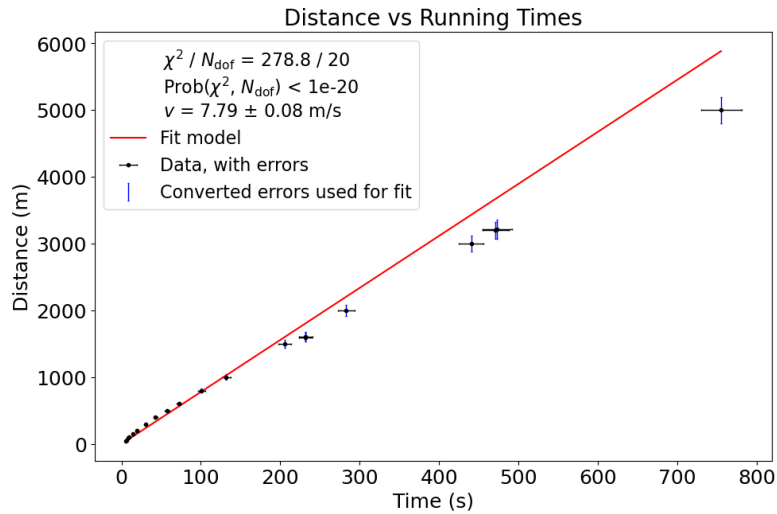


Figure 22: Linear function $f_0(t) = vt$ fitted to distances vs running time with converted errors from time to distance.

5.1.2

For modeling the running times as a function of distance I tried a linear, quadratic, exponential, and power function. The χ^2 , N_{dof} , and p -value for fit model is listed in table 10. Only the power function has a p -value that is sufficiently large to not reject the hypothesis that it is a good model. The fit is shown in fig. 23.

Model	χ^2	N_{dof}	p
Exponential	4762	19	$< 10^{-20}$
Linear	149	19	$< 10^{-20}$
Quadratic	61.3	18	1×10^{-6}
Power	27.2	19	0.1

Table 10: Results from different fits in 5.1.2.

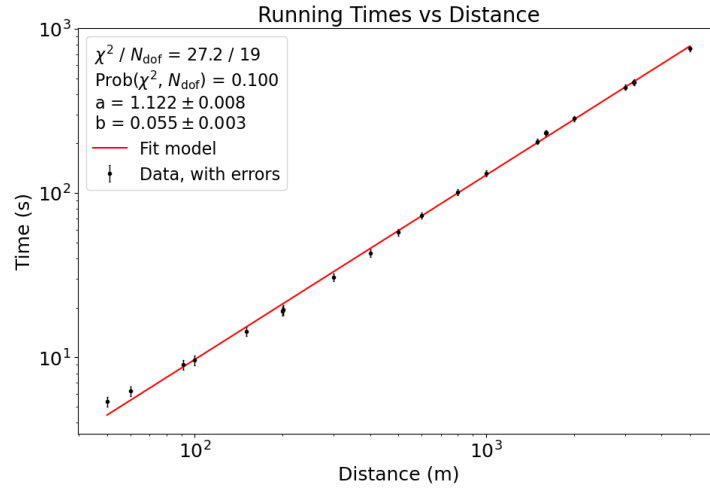


Figure 23: Power law fit to running time as a function of distance. Double-log axes are used.

To find out which records are most likely to improve I calculated the residual of each point and normalized by dividing by its uncertainty so it corresponds to how many sigmas away from the model it is. Those more than 2 sigmas away (red) are the most likely to be improved. Those between 1 and 2 sigmas away (blue) are also likely to be improved since they deviate more than 1 uncertainty from the model. These marked distances are listed in table 11.

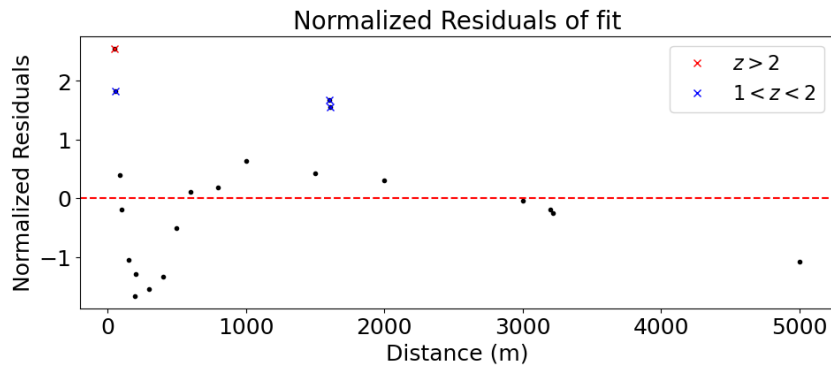


Figure 24: Residuals of data points divided by their uncertainty. Those with $z > 2$ are marked with red and those with $1 < z < 2$ are marked with blue.

Distance	Residual (s)	Normalized residual (z)
50 <i>m</i>	0.94	2.55
60 <i>m</i>	0.78	1.82
1600 <i>m</i>	13.90	1.67
1609.3 <i>m</i>	12.78	1.56

Table 11: Results from normalized residuals in 5.1.2 showing distances which time is the most likely to be improved.

5.2

5.2.1

The distribution of frequency measurements is shown in fig. 25. I fitted the two major peaks with the function

$$f(x) = N_{\text{gauss}} G(x; \mu, \sigma) + N_{\text{exp}} \frac{1}{\tau} \exp\left(\frac{-(x - x_0)}{\tau}\right), \quad (24)$$

where $G(x; \mu, \sigma)$ is a Gauss. The two local fits are shown in fig. 26a and fig. 26b. For both fits I used χ^2 fit since all bins have over 50 entries. For the upper peak $p = 0.188$ while for the lower $p < 10^{-10}$. The hypothesis that the model describes the data correctly cannot be rejected for the upper peak, but it can for the lower.

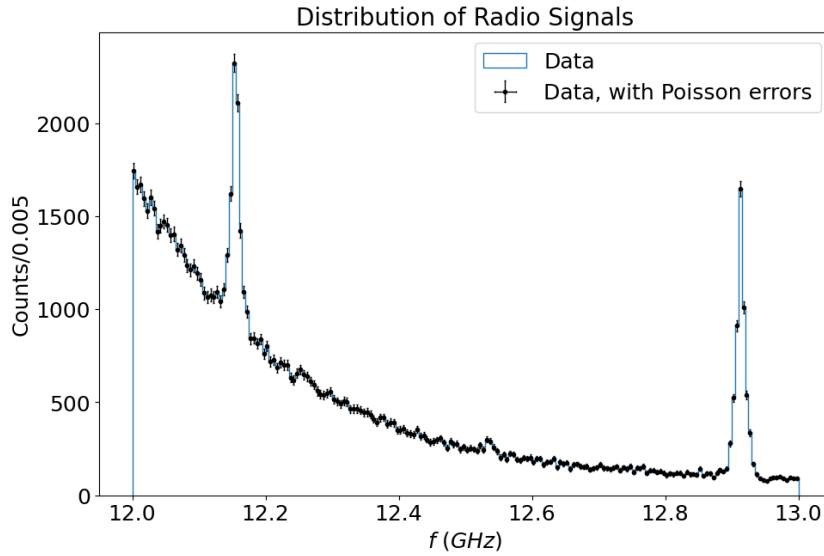
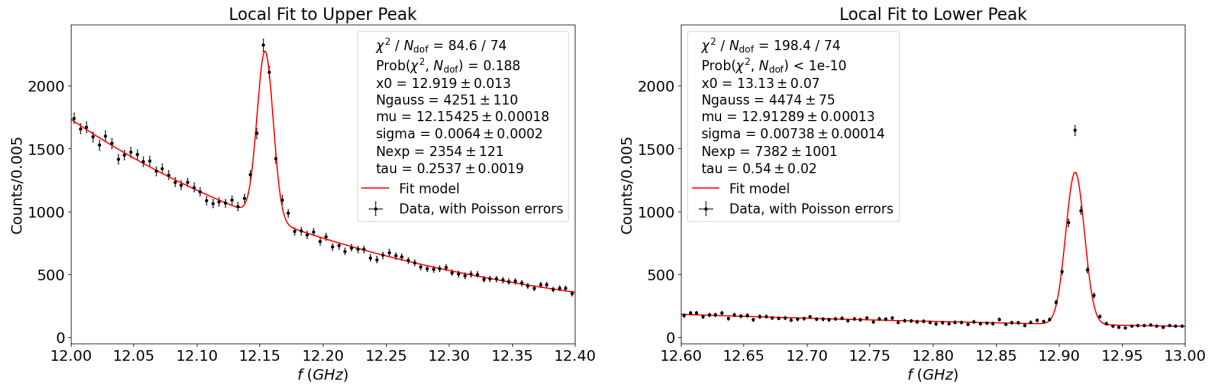


Figure 25: Distribution of frequency measurements.



(a) Upper peak fitted with shifted exponential + gauss PDF. $p = 0.188$ indicates a good fit. (b) Lower peak fitted with shifted exponential + gauss PDF. $p < 10^{-10}$ indicates a poor fit.

Figure 26: Local fits to the peaks in the frequency measurements.

I also fitted the lower peak with a exponential + Cauchy. Here $p = 0.00002$ which still means that it is a poor fit, but it is much better than the previous with gauss. The fit is shown in fig. 27

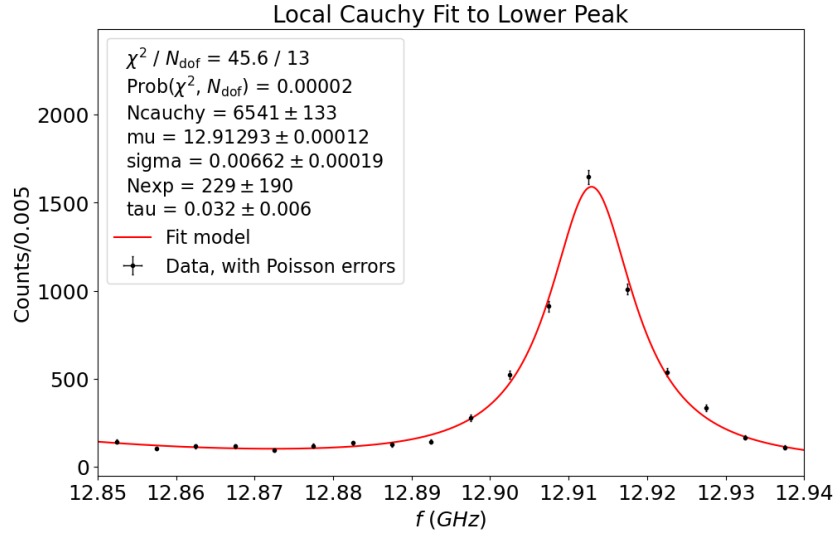


Figure 27: Lower peak fitted with shifted exponential + Cauchy PDF. $p = 0.00002$ indicates a poor fit, but much better than for the exponential + gauss.

5.2.2

The upper peak is consistent with a gauss $p = 0.188$ while the lower peak is not $p < 10^{-10}$. To determine whether they are consistent with each other I compared the fitted parameter N_{gauss} which yielded

$$z = 1.68 \text{ and } p = 0.09, \quad (25)$$

which is not a sufficiently low (two-sided) p -value to reject the hypothesis that the two peaks are consistent (equal amount of measurements in) with each other.

5.2.3

In order to remove some of the noise I first calculate how many sigmas away every point of E and D were from their respective means. I called these variables Z_E and

Z_D . I then calculated

$$S = Z_E^2 + Z_D^2, \quad (26)$$

which I used as a cutoff parameter to remove points that had far away E and D values (outliers). If I made the cutoff at too low a value of S the peaks would disappear. I decided to make the cut at $S = 1.5$ since this gave much better χ^2 for the fits compared to before while still preserving the peaks. This meant removing 48414 data points equivalent to 0.48% of the measurements. The distribution of S is shown in fig. 28a and the filtered frequencies in fig. 28b.

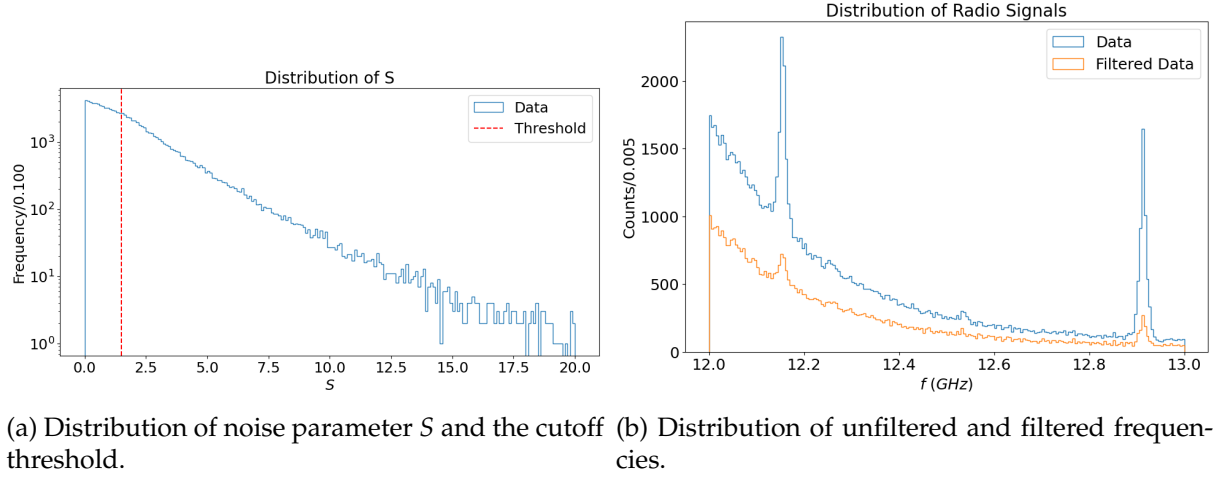


Figure 28: Noise parameter and filtered frequency.

Refitting the two peaks yields $p = 0.657$ while for the lower $p = 0.019$. This is much better especially for the lower peak. The hypothesis that the model describes the data correctly still cannot be rejected for the upper peak, and for the lower peak it can be accepted at a 99% CI, but still not at 95%.

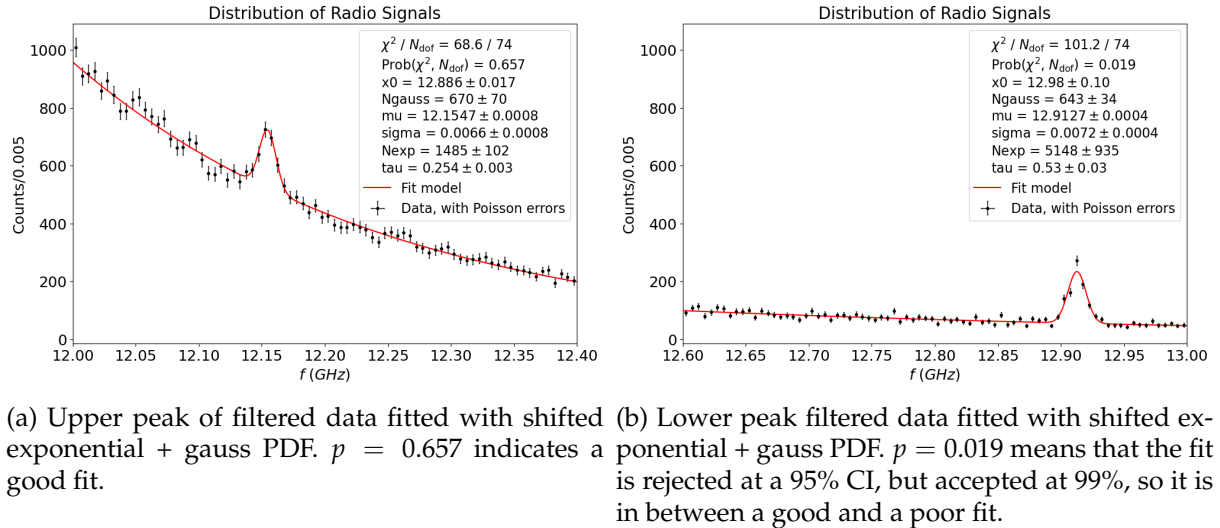


Figure 29: Local fits to the peaks in the frequency measurements.

I also fitted the filtered lower peak with a constant + Cauchy. Here $p = 0.244$ which is not sufficient to reject the hypothesis that the model describes the data well. However, the fit is fitted to a very narrow part of the data. Here the cutoff value used for S is 3. The fit is shown in fig. 30.

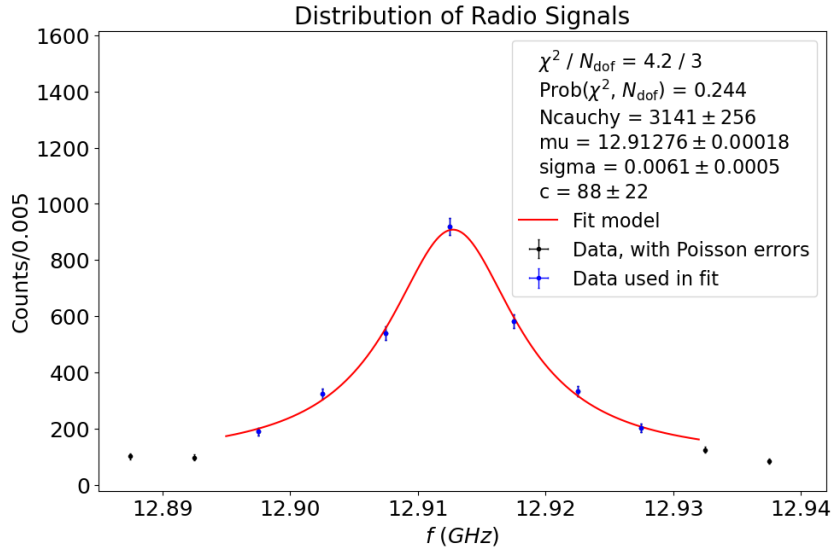


Figure 30: Lower peak fitted with a constant + Cauchy PDF. $p = 0.244$ indicates a good fit. Few data points are used (marked with blue) and the cutoff value for S is 3.

5.2.4

Fitting the data for $f \in [12.4, 12.7]$ with the same function as with the two major peaks yields a fit with $p = 0.923$ indicating that there is a peak midway between the two large peaks. The fit is shown in fig. 31.

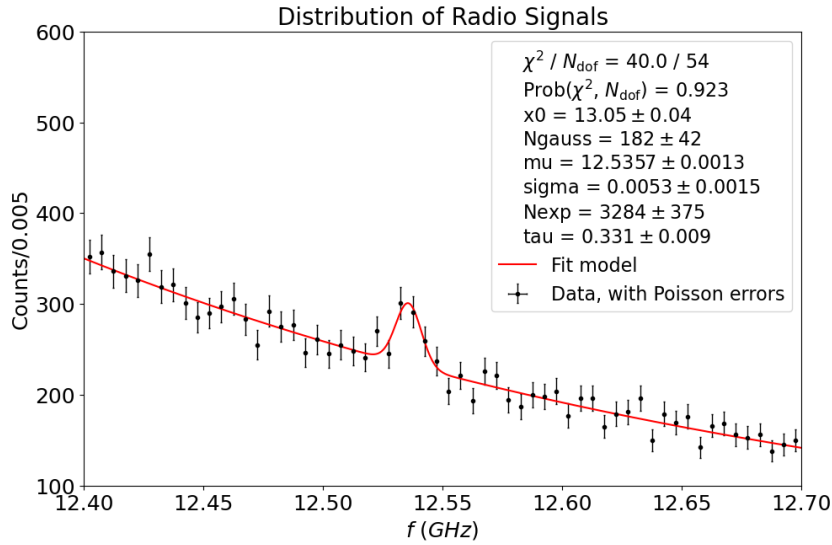


Figure 31: Midway peak fitted with shifted exponential + gauss PDF. $p = 0.923$ indicating a peak and a great fit.